



**CEBU INSTITUTE OF TECHNOLOGY**  
**U N I V E R S I T Y**

# IT342-Section SYSTEMS INTEGRATION AND ARCHITECTURE 1

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## FUNCTIONAL REQUIREMENTS SPECIFICATION (FRS)

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Project Title: Mini App – User Registration & Authentication

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## 1. Introduction

### 1.1. Purpose

The purpose of this document is to describe the design and system flow of a Mini Application that supports user registration, login, logout, and access to protected pages. This document is intended for students, instructors, and developers who will design, review, and implement the system. It serves as a reference for the system's functionality and structure before actual coding is done.

### 1.2. Scope

The system will allow users to create an account, log in, view a protected profile/dashboard, and log out. The system will restrict access to protected pages when the user is not authenticated. This documentation covers only the system design and diagrams (ERD, Use Case, Activity, Class, and Sequence diagrams). Actual coding and implementation are outside the scope of this document.

### 1.3. Definitions, Acronyms, and Abbreviations

- User – A person who uses the system
- Guest User – A user who is not logged in
- Authenticated User – A user who has successfully logged in
- ERD – Entity Relationship Diagram
- API – Application Programming Interface
- UI – User Interface
- JWT – JSON Web Token (used for authentication tokens)
- CRUD – Create, Read, Update, Delete

## 2. Overall Description

### 2.1. System Perspective

The system is a simple authentication module within a web application. It follows a client-server architecture where the frontend (React UI) communicates with a backend API (Spring Boot), which interacts with a database to store and retrieve user information. The authentication system may later be integrated into a larger application such as an e-commerce site, student portal, or management system.

### 2.2. User Classes and Characteristics

Guest User - Can access public pages and register or log in

Authenticated User - Can view profile/dashboard and access protected pages

### 2.3. Operating Environment

- Frontend: React
- Backend: Spring Boot (Java)
- Database: MySQL / PostgreSQL

- Tools:
  - IDE (IntelliJ / VS Code)
  - Browser (Chrome, Firefox)
  - API testing tool (Postman)
  - Platform: Web-based application running on a local or cloud server

## 2.4. Assumptions and Dependencies

### Assumptions:

- Users have access to the internet and a modern web browser
- Users provide valid registration information
- Passwords are stored securely (hashed)

### Dependencies:

- Database availability
- Backend API running properly
- Token provider or authentication library (e.g., JWT)
- Password encoding library

## 3. System Features and Functional Requirements

### 3.1. Feature 1: User Registration

Description: Allows a guest user to create a new account in the system by providing required details.

Functional Requirements:

- The system shall allow users to register using a username/email and password
- The system shall validate user input (e.g., required fields, password length)
- The system shall store user data securely in the database

### 3.2. Feature 2: User Login & Authentication

Description: Allows registered users to log in and gain access to protected features.

Functional Requirements:

- The system shall authenticate users using valid credentials
- The system shall generate an authentication token upon successful login
- The system shall restrict access to protected pages for unauthenticated users

### 3.3. Feature 3: View Profile/Dashboard

Description: Allows registered users to log in and gain access to protected features.

Functional Requirements:

- The system shall display user-specific information after login
- The system shall prevent guest users from accessing the dashboard
- The system shall retrieve user data from the database

### 3.4. Feature 2: User Logout

Description: Allows authenticated users to log out of the system and end their session.

Functional Requirements:

- The system shall invalidate the user session/token upon logout
- The system shall redirect the user to the login or home page
- The system shall prevent access to protected pages after logout

## 4. Non-Functional Requirements

### Security:

- Passwords must be encrypted/hashed
- Authentication tokens must be securely generated and validated
- Protected routes must require authentication

### Performance:

- Login and registration should respond within 2 seconds under normal load
- Usability:
- The UI should be simple and user-friendly
- Error messages should be clear and understandable

### Reliability:

- The system should handle invalid login attempts gracefully
- The system should maintain session consistency

### Scalability:

- The design should allow future expansion (e.g., roles, permissions, password reset)

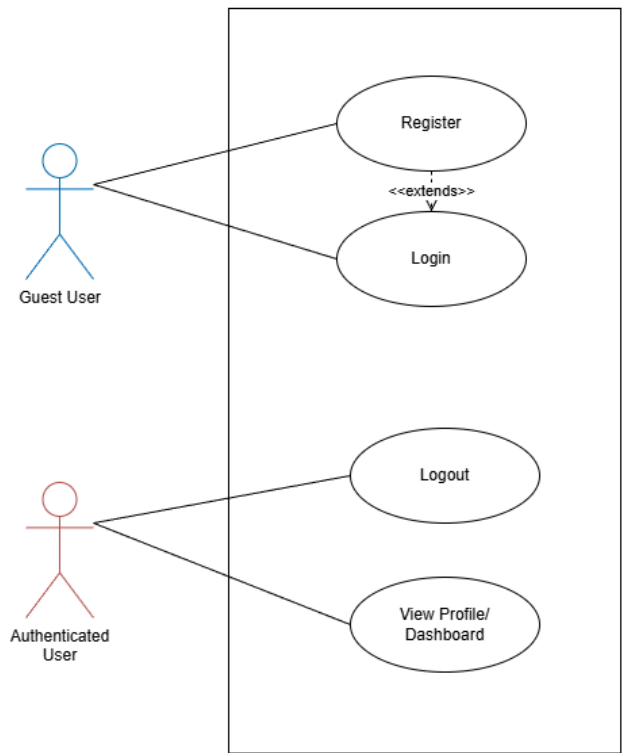
# 5. System Models (Diagrams)

## 5.1. ERD

| USERS                         |
|-------------------------------|
| <b>user_id: BIGINT (PK)</b>   |
| username: VARCHAR(100) UNIQUE |
| email: VARCHAR(255) UNIQUE    |
| password_hash: VARCHAR(255)   |
| first_name: VARCHAR(50)       |
| last_name: VARCHAR(50)        |
| is_active: BOOLEAN            |
| created_at: TIMESTAMP         |
| updated_at: TIMESTAMP         |

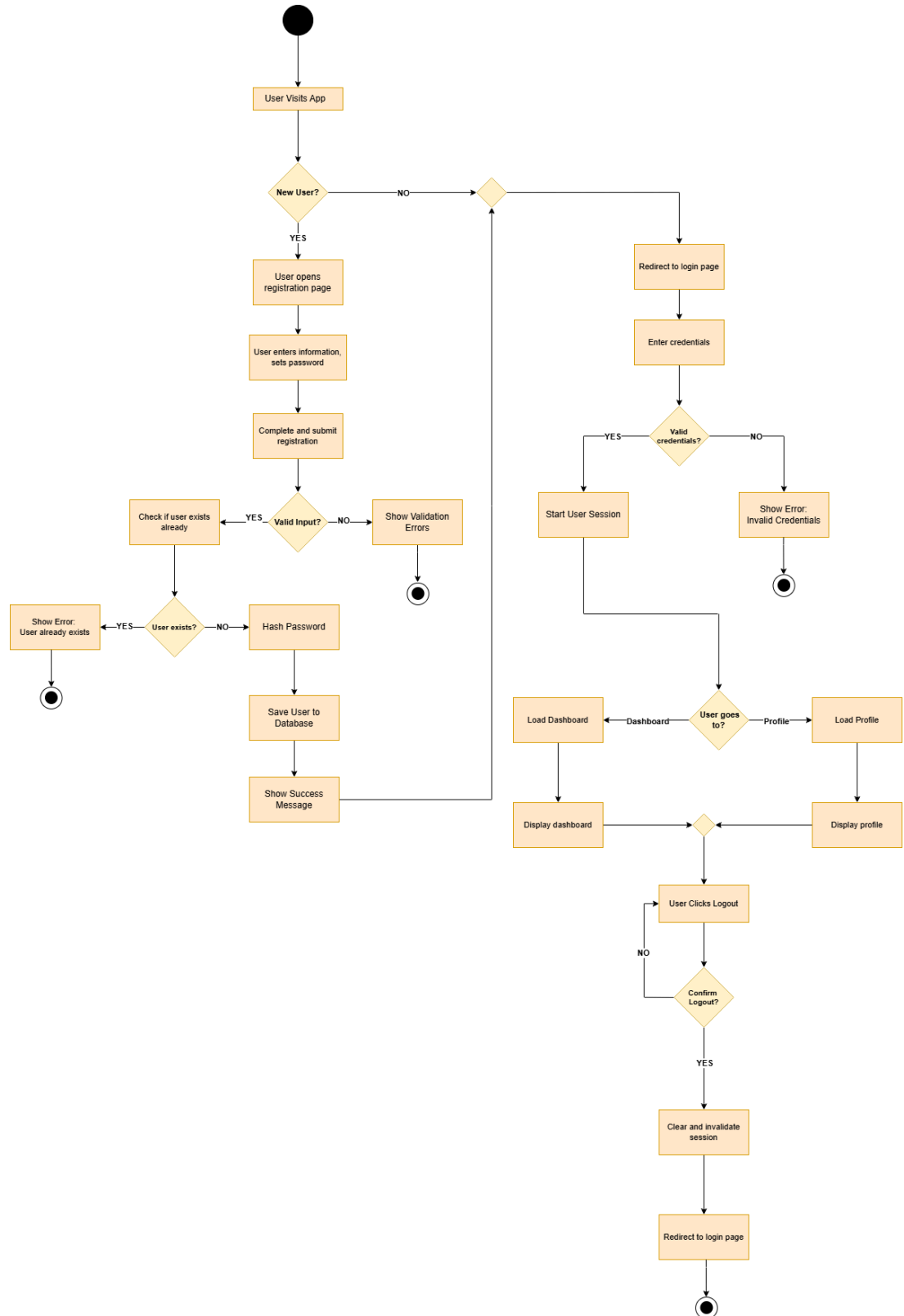
## 5.2. Use Case Diagram

Use Case Diagram



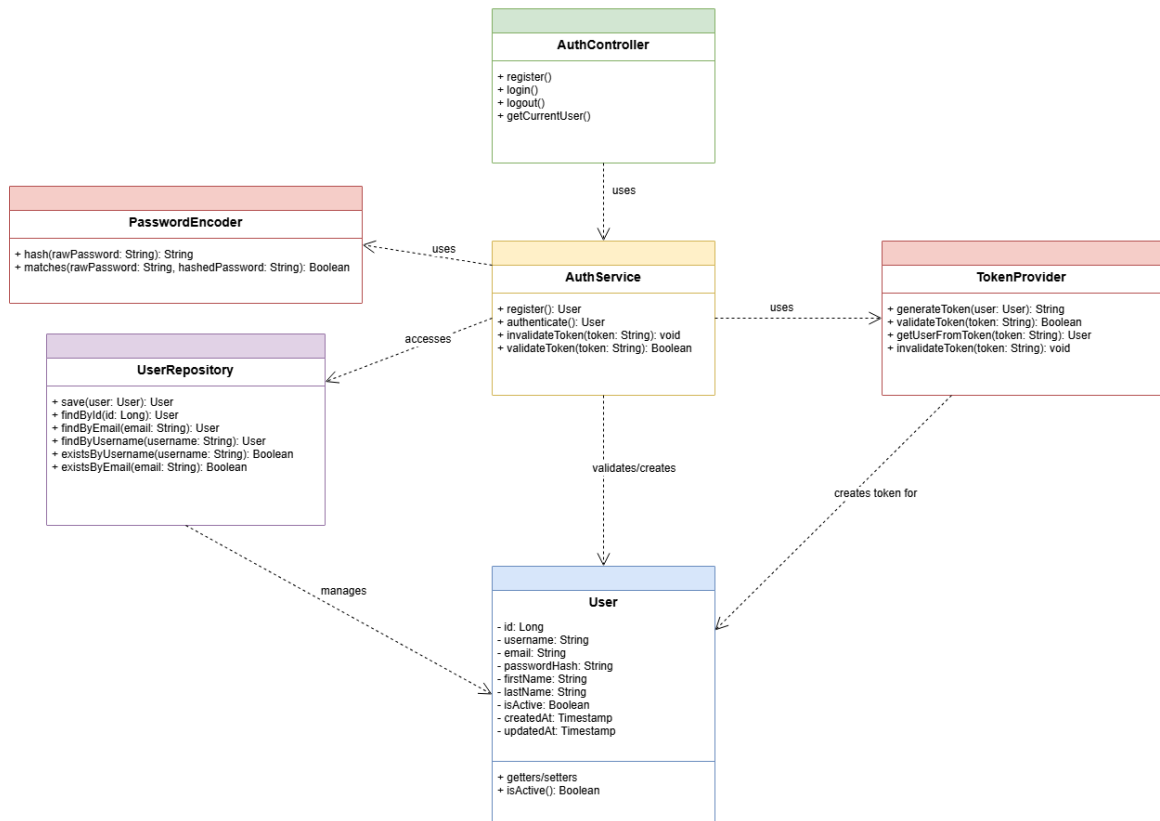
### 5.3. Activity Diagram

Activity Diagram - Registration, Login, and Logout Flow



## 5.4. Class Diagram

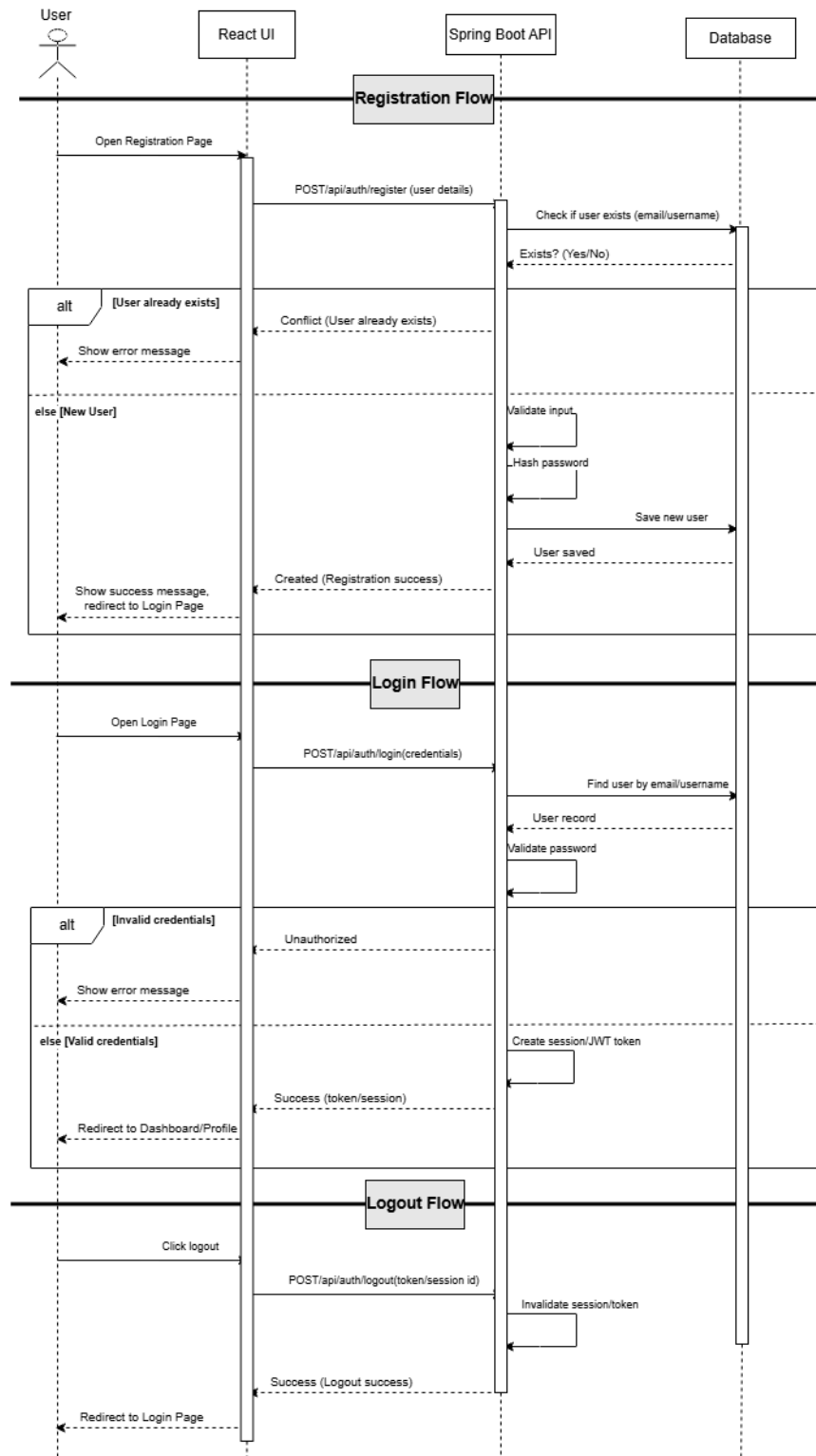
Class Diagram - Authentication System Backend





## 5.5. Sequence Diagram

Mini App Register, Login, and Logout Sequence Diagram

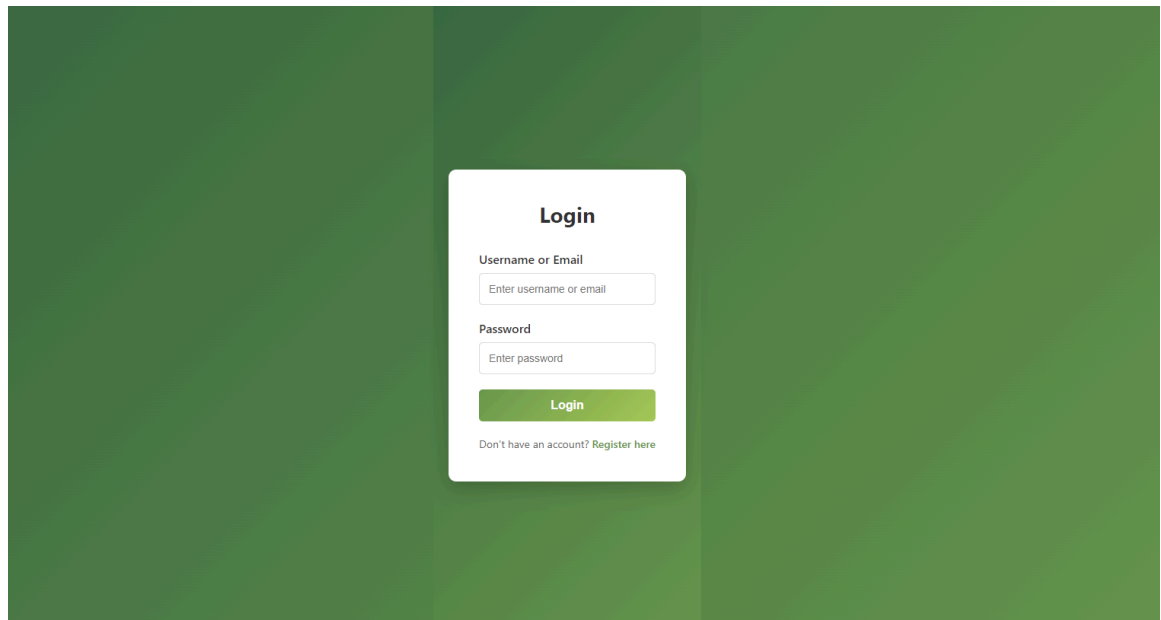


## 6. Appendices

N/A

## 7. Screenshots of the Web UI

### a. Login Page (Index)



A screenshot of a login page with a dark green background. A white login form is centered. The form has a title 'Login', a 'Username or Email' field with placeholder text 'Enter username or email', a 'Password' field with placeholder text 'Enter password', a green 'Login' button, and a link 'Don't have an account? Register here'.

**Login**

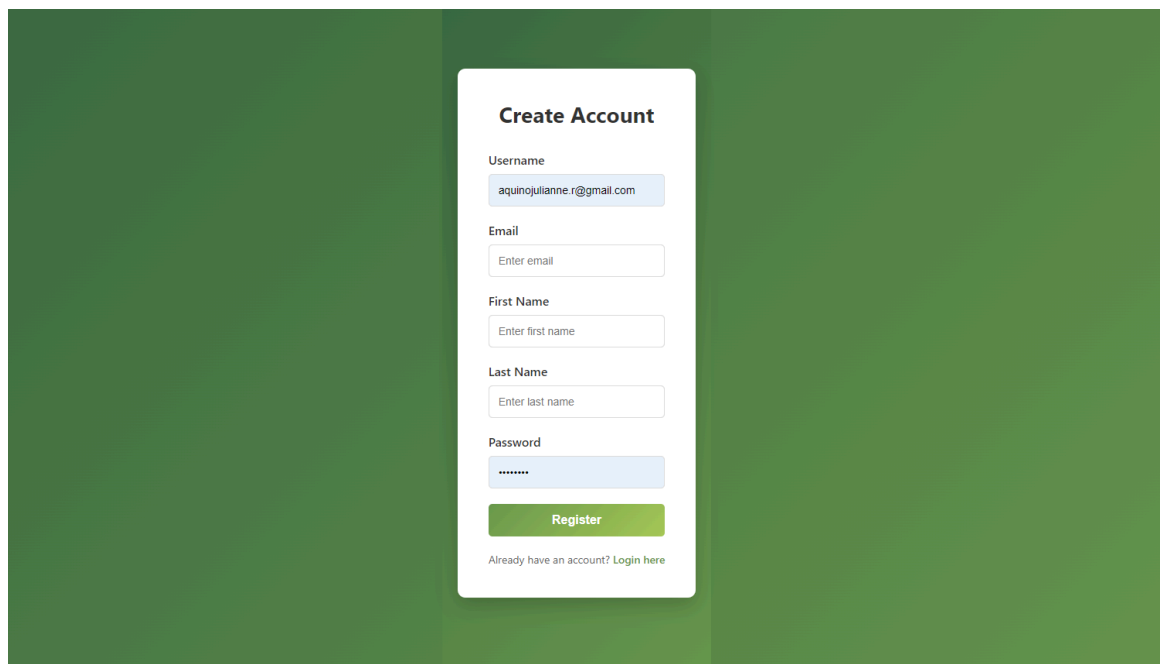
Username or Email  
Enter username or email

Password  
Enter password

Login

Don't have an account? [Register here](#)

### b. Register Page



A screenshot of a register page with a dark green background. A white registration form is centered. The form has a title 'Create Account', fields for 'Username' (with value 'aquinojulianne.r@gmail.com'), 'Email' (with placeholder 'Enter email'), 'First Name' (with placeholder 'Enter first name'), 'Last Name' (with placeholder 'Enter last name'), and 'Password' (with placeholder '\*\*\*\*\*'). It includes a green 'Register' button and a link 'Already have an account? Login here'.

**Create Account**

Username  
aquinojulianne.r@gmail.com

Email  
Enter email

First Name  
Enter first name

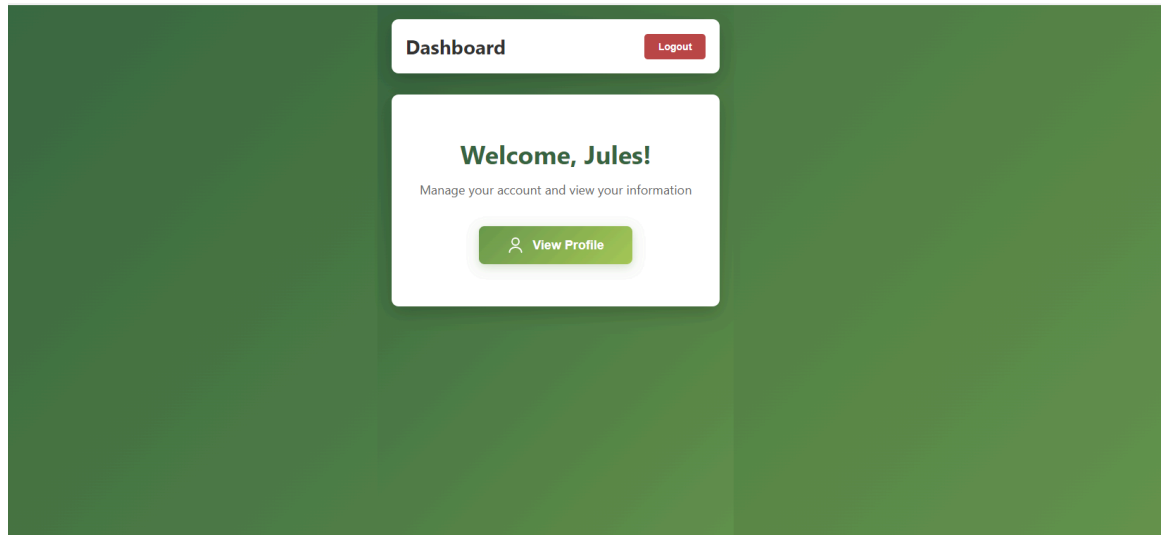
Last Name  
Enter last name

Password  
\*\*\*\*\*

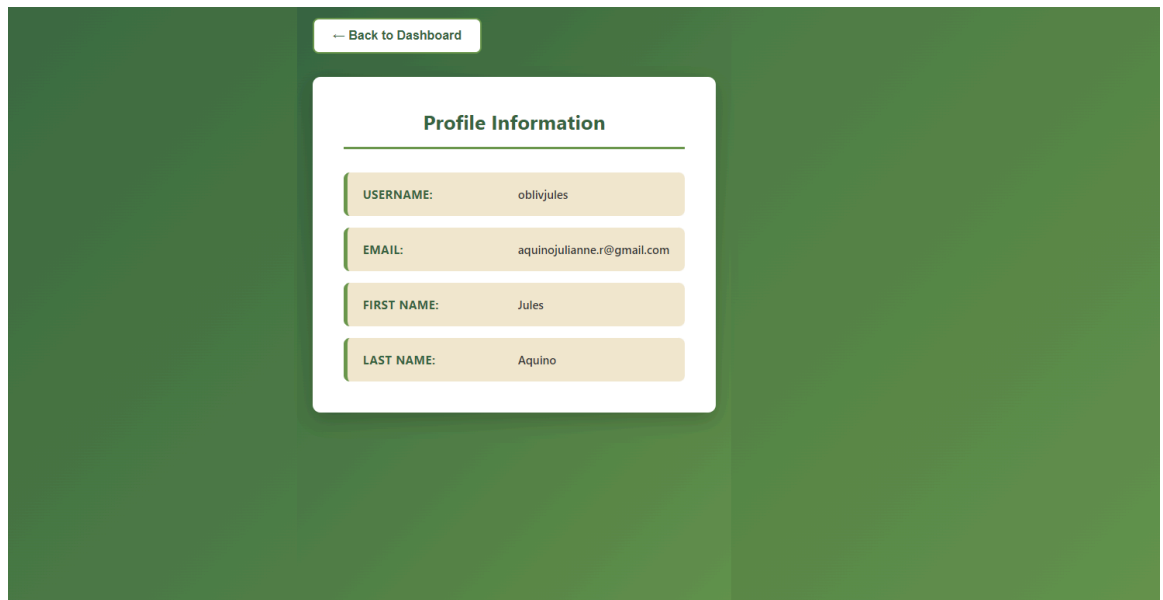
Register

Already have an account? [Login here](#)

c. Dashboard



d. Profile



e. Logout Confirmation

